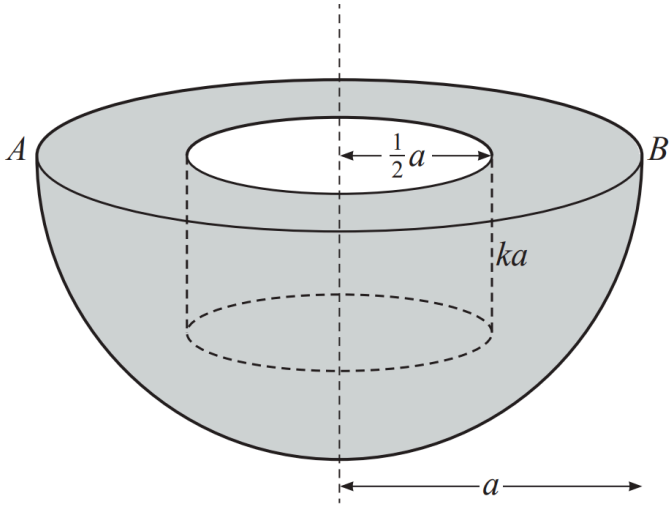
(FM) 2. Equilibrium of a rigid body HW

NO. 1



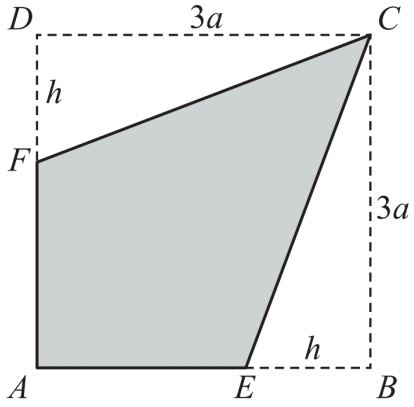
An object is formed by removing a solid cylinder, of height ka and radius $\frac{1}{2}a$, from a uniform solid hemisphere of radius a. The axes of symmetry of the hemisphere and the cylinder coincide and one circular face of the cylinder coincides with the plane face of the hemisphere. AB is a diameter of the circular face of the hemisphere (see diagram).

a. Show that the distance of the centre of mass of the object from AB is $\frac{3a(2-k^2)}{2(8-3k)}$. [4]

When the object is freely suspended from the point A, the line AB makes an angle θ with the downward vertical, where $\tan\theta = \frac{7}{18}$.

b. Find the possible values of k. [3]

NO. 2



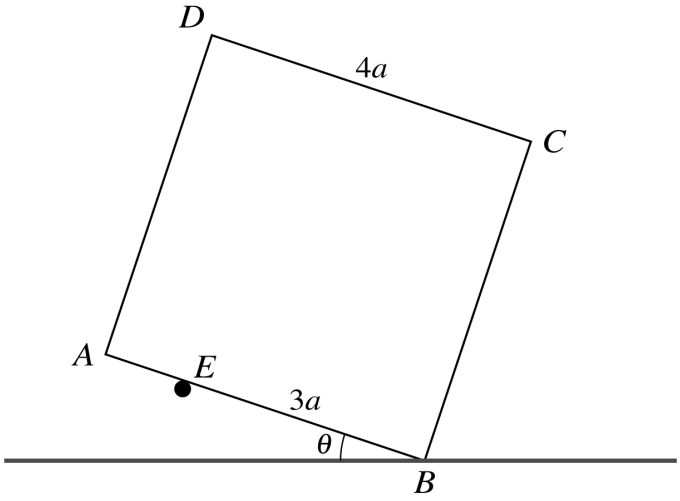
A uniform lamina $AECF$ is formed by removing two identical triangles BCE and CDF from a square lamina $ABCD$. The square has side $3a$ and $EB = DF = h$ (see diagram).

a. Find the distance of the centre of mass of the lamina $AECF$ from AD and from AB, giving your answers in terms of a and h. [5]

The lamina $AECF$ is placed vertically on its edge AE on a horizontal plane.

b. Find, in terms of a, the set of values of h for which the lamina remains in equilibrium. [3]

NO. 3

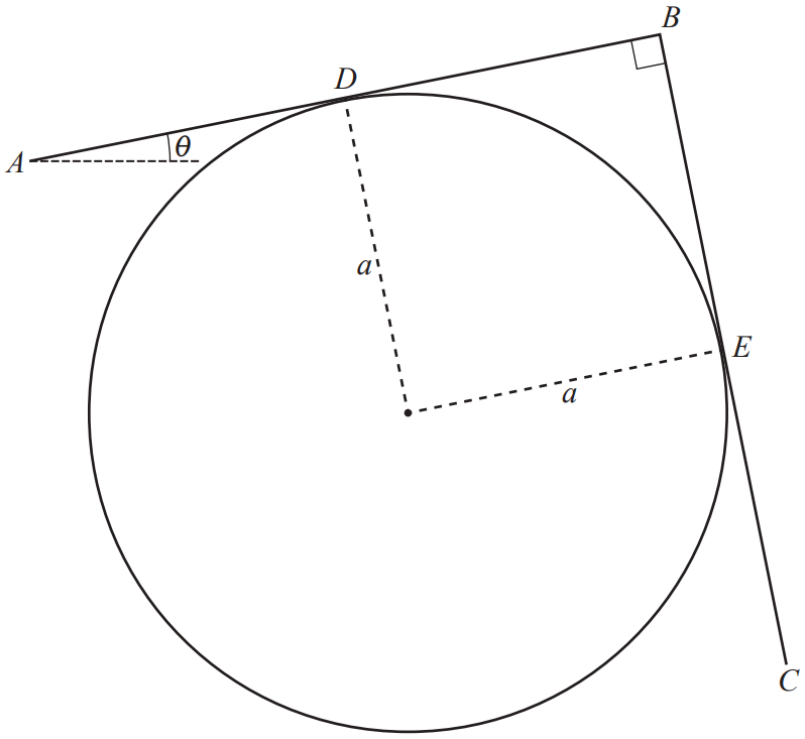


A uniform square lamina $ABCD$ of side $4a$ and weight W rests in a vertical plane with the edge AB inclined at an angle θ to the horizontal, where $\tan\theta = \frac{1}{3}$. The vertex B is in contact with a rough horizontal surface for which the coefficient of friction is μ. The lamina is supported by a smooth peg at the point E on AB, where $BE = 3a$ (see diagram).

i. Find expressions in terms of W for the normal reaction forces at E and B. [5]

ii. Given that the lamina is about to slip, find the value of μ. [3]

NO. 4



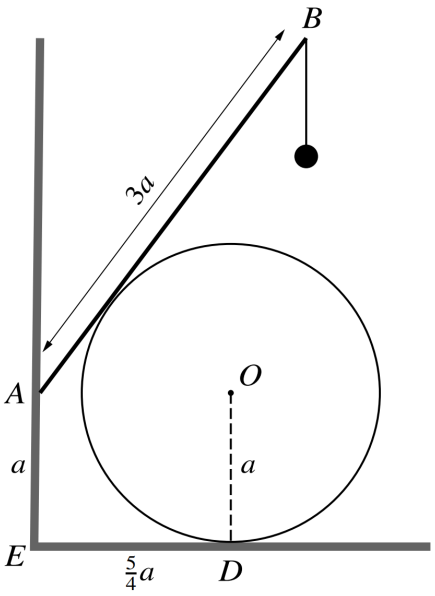
A uniform cylinder with a rough surface and of radius a is fixed with its axis horizontal. Two identical uniform rods AB and BC, each of weight W and length $2a$, are rigidly joined at B with AB perpendicular to BC. The rods rest on the cylinder in a vertical plane perpendicular to the axis of the cylinder with AB at an angle θ to the horizontal. D and E are the midpoints of AB and BC respectively and also the points of contact of the rods with the cylinder (see diagram). The rods are about to slip in a clockwise direction. The coefficient of friction between each rod and the cylinder is μ.

The normal reaction between AB and the cylinder is R and the normal reaction between BC and the cylinder is N.

a. Find the ratio $R : N$ in terms of μ. [6]

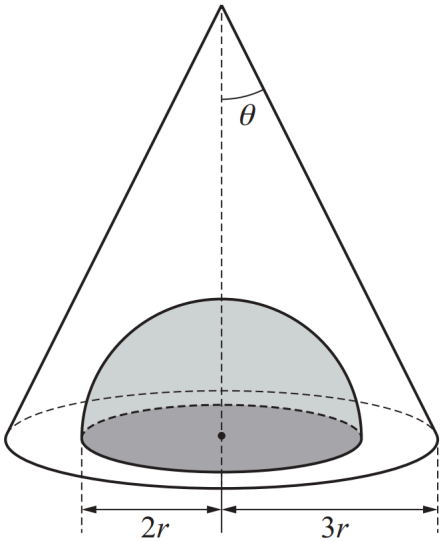
b. Given that $\mu = \frac{1}{3}$, find the value of $\tan\theta$. [3]

NO. 5



A uniform smooth disc with centre O and radius a is fixed at the point D on a horizontal surface. A uniform rod of length $3a$ and weight W rests on the disc with its end A in contact with a rough vertical wall. The rod and the disc lie in a vertical plane that is perpendicular to the wall. The wall meets the horizontal surface at the point E such that $AE = a$ and $ED = \frac{5}{4}a$. A particle of weight kW is hung from the rod at B (see diagram). The coefficient of friction between the rod and the wall is $\frac{1}{8}$ and the system is in limiting equilibrium. Find the value of k. [8]

NO. 6



An object is formed by removing a solid hemisphere, radius $2r$, from a uniform solid cone, radius $3r$ and semi-vertical angle θ, where $\tan\theta = \frac{1}{2}$. The axes of symmetry of the cone and the hemisphere coincide. The base of the cone and the base of the hemisphere are in the same plane as each other (see diagram).

a. Find, in terms of r, the distance of the centre of mass of the object from its base. [4]

The object is placed such that its circular base makes contact with a rough plane which is inclined to the horizontal at an angle α. The object is on the point of toppling. The plane is sufficiently rough to prevent sliding.

b. Find the value of α. [3]